Keep the idea. Rebuild the execution.

Calibrated Paper was the right direction. Its weaknesses were in craft: identity, type rhythm, density and a few broken layouts.

Keep

- Ink, paper, green, gold palette and their single meanings
- Solid measured, dashed experimental, hatched uncertainty
- Plex Sans and Plex Mono for text and values
- Status as icon plus word plus border style
- Research notices on every reading

Improve

- Logo: a drawn wordmark replaces font text plus icon
- One type scale shared by mobile and web, 4px rhythm
- Web data text up from 12 to 14px, table rows 44px
- Separate navigation, analysis and methodology regions
- Scanner: real camera layout with honest feedback

Remove

- Strip symbol: read as a battery at small sizes
- Bricolage display face: a third family with no job
- Wide tracked mono CHECK and uppercase eyebrows
- Decorative ruled boxes and placeholder photo tiles
- Hour timer as a hero number on vendor screens

Defects found when rendering v1

Where	Problem	v2 resolution
Master moodboard	Five panels fell out of the grid; fonts fell back to a serif; a large empty band	Brand kit rebuilt on a fixed 12 column grid
Strip results (p.6)	Collision reported in the PDF export; the expanding disclosure and tight label stack are the likely cause	Explanation is a fixed block; labels have their own rows
Logo	Symbol unreadable below 24px; MANTIKA and CHECK unrelated	Gauge A wordmark; A is the icon
Web tables	12px captions and mono, 36px rows, notes below the fold	14px minimum, 44px rows, method panel

Two answers, one chosen

Both are vector drawn. A makes the letters do the work. B adds a symbol. A wins because its mark is already inside the name.

A · Editorial wordmark

Selected

MANTIKA CHECK

The gauge A

The crossbar stops short of the left stroke and fuses into the right, like a needle reading a level. Gold in colour, ink in mono.

Two weights, one height

MANTIKA is drawn at a 17 unit stroke, CHECK at 10, on a shared 100 unit cap height. Kitchen word, lab word, one line.

Flat terminals

Every stroke is cut square to the cap and base line, with optical kerning on MA, AN and NT.

B · Measurement identity

MANTIKA Check

Three tonal steps for the three strip bands. Clear at 16px and a strong app tile, but at small sizes it reads as a signal or volume meter, and the wordmark beside it is ordinary set type.

Criterion	A · Editorial wordmark	B · Measurement identity
Proprietary feel	Drawn letters, one recognisable detail	Symbol is generic; type is a font
16px recognition	Gauge A with gold bar stays distinct	Clear, but confusable with meters
Fit with the science	A reading, not a verdict	Implies a graded scale we have not validated

03 · Brand kit · SVG files in design/refined/brand

Lockups, icon and rules

MANTIKA CHECK

Primary horizontal · ink and gold on paper or white

MANTIKA CHECK

Reversed · paper and light gold on ink

MANTIKA CHECK

Monochrome · print, stamps, one colour forms

MANTIKA
CHECK

Compact · stacked, CHECK at half size



App icon · the gauge A alone. Ink tile is default; paper tile for light launchers.

Small size legibility, actual pixels



Below 120px wide, CHECK's thin strokes fall under one pixel: use the icon instead.



- Clear space x = cap height of the wordmark on every side
- Minimum: lockup 120px, compact 72px, icon 16px
- Never re-set CHECK in a font, stretch, or recolour the A bar in anything but gold or the ink

Design system

Two families, one type scale, a 4px grid, two radii. Ratios are measured WCAG values on white unless noted.

Colour

ink
#14213D · 15.9

paper
#F6F3EC · ground

green
#2F4A39 · 9.7

gold
#B07A2A · 3.7 graphic

Role	Token	Value	Ratio
Body text	text.primary	#14213D	15.9
Secondary text	text.secondary	#374151	10.3
Lightest allowed text	text.muted	#4B5563	7.6
Experimental text	exp.text / exp.bg	#6B4A12 / #F3E6CC	6.5 on bg
Error	error.text / error.bg	#8E2F1C / #FBEFE9	7.2 on bg
Done, synced	ok.text / ok.bg	#2F4A39 / #EAF0EA	8.4 on bg
Control outline	border.control	#8C8574	3.3 on paper
Divider (decorative)	border.rule	#DDD6C6	n/a

Type scale · one scale, both platforms

Display 32 — 600 · 1.15 · web titles

Title 24 — 600 · 1.2 · screen titles

Heading 20 — 600 · 1.3 · sections

Body 17 mobile — 400 · 1.5

Body 15 web, tables — 400 · 1.5

Label 15 — 600 · fields, buttons

Caption 14 · sources, hints — 400 · minimum text

24.6 % · 190 °C — Plex Mono · values only

No uppercase eyebrows, no tracking above 0. Mono is reserved for numbers, units, IDs and file names.

Space, grid, radius

4 8 12 16 24 32 48

- Mobile 320 to 430: 4 col, 16px margin, 12px gutter
- Tablet 768: 8 col, 32px margin, rail collapses to top bar
- Desktop 1024 to 1440: 12 col, 40px margin, 24px gutter, max 1280
- Radius 4: badges, inputs, tags. Radius 8: buttons, panels
- Borders 1px; 2px only for focus and selected

Controls · mobile 56 / 48, web 40

Primary 56

Secondary 48

Tertiary text

Web 40 · focus ring

Disabled, with reason

Oil type: Palm oil

Temperature, if known: 1750

Check the number. Frying oil is usually 150 to 200 °C.

Readings | Oil sessions

1 | 2 | 3

− 175 °C +

Status vocabulary · shape + word + colour

✓ Sent · □ Saved on phone · ◌ Waiting to send · ⚠ Not sent · ··· Experimental · – Not available

✓ Reading saved on this phone. Undo

Toasts last 6 seconds, never carry the only copy of an error, and stay until dismissed when they hold an action.

Chart tokens

○ 160 °C	□ 175 °C	△ 190 °C	◇ 205 °C	× 220 °C
#7C8AA3	#5D6B85	#3E4E6C	#253047	#14213D

- Temperature is ordered, so one ink ramp by lightness; every series also has its own marker. All ≥ 3.4:1 on white
- Measured: solid 2px line, filled marker. Experimental: gold #B07A2A 2px dashed 6 4, hollow marker, "Experimental" label at line end
- Uncertainty: gold 45° hatch, never a gradient. Benchmark: 1.5px dotted, labelled in text
- Axes 1px ink, grid #EAE4D6, ticks 13px mono; direct labels before legends
- Every chart: title, unit, source line, "Show table" toggle. Under 600px charts become tables

System states

No readings yet
Your first strip scan will appear here.

Loading readings...

No internet. Saving on phone.

Photo too dark. Move to shade or turn on the light.

Accessibility specification

1. Text 14px minimum, 17px mobile body; no text lighter than #4B5563
2. Targets 48px mobile (56 primary), 40px web with 8px spacing
3. 2px gold focus ring with 2px offset, visible on ink and paper
4. Status always icon + word; colour third
5. Layouts reflow at 320px and 200% zoom; labels wrap, never truncate
6. Strings sized for Filipino at +35% length
7. Motion ≤ 200ms, none under reduced motion

MANTIKA CHECK

✓ All sent

Stall V-07

Thursday, 23 October

Latest strip reading



Band 2 of 3

Today, 10:42 AM · you confirmed it

··· Experimental reading

›

Today's oil

Palm oil · Kwek-kwek since 7:10 AM

Heating recorded	Fresh oil added
3 h 20 m	1 time

Update oil log

Recent

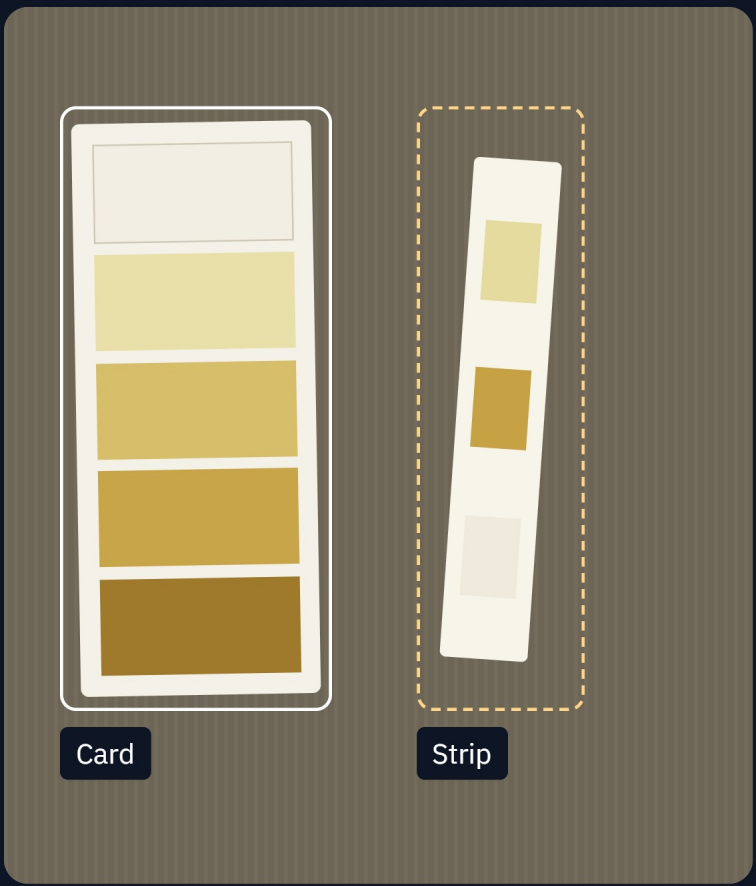
Strip scanned	10:42
Added fresh oil	08:15
Started oil log · Palm oil	07:10

Scan a test strip

Step 2 of 3

Line up the strip and the colour card

Lay both flat. Hold the phone about one hand above them.



Light
✓ Good

Focus
✓ Sharp

Strip
△ Tilted

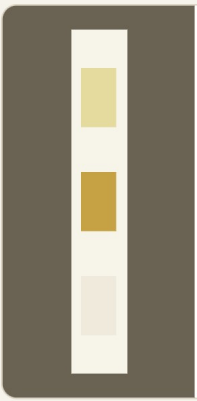
These checks look at photo quality only, not at the oil.

Help

Light

Check the reading

Step 3 of 3



Phone reading

Band 2 of 3

Band 1	Changed
Band 2	**Changed**
Band 3	No change

What this means

Each band changes colour when certain compounds from reused oil build up. Two changed bands means more build up than one. It is not a safety result.

Look at the strip. Which band is it?

Band 1 | **Band 2** | Band 3

If you pick a different band, both answers are kept for the research team.

Experimental. The phone reading is still being compared with people's readings. It is not a lab test.

Retake

Save reading

Oil log

Past days

1 · What oil and food?

Oil

Palm | Coconut | Soybean | Not sure

Food

Kwek-kwek | Fish balls | Banana cue | Other

2 · Heating

3 h 20 m

● **Frying now · tap to pause**

Oil temperature, if you know it

− 175 °C + | Don't know

3 · Oil changes today

+ Added fresh oil | **Replaced all oil**

Fresh oil added 1 time, at 8:15 AM.

Done for today

History

Compare two

All | Strip readings | Oil logs | Needs y

1 reading not sent. It is still saved on this phone.

Send again

Today · Thu 23 Oct



Band 2 of 3
10:42 · after 3 h 20 m heating

Waiting

?

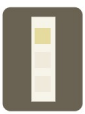
Reading not finished
08:05 · photo missing

Finish

Oil log started · Palm
07:10 · fresh oil added 08:15

Sent

Yesterday · Wed 22 Oct



Band 1 of 3
16:10 · after 5 h heating

Not sent



No band changed
09:30 · fresh oil

Sent

Home | Oil log | History

Mobile states · one pattern, ten situations

Icon + plain title + what happens next + one action

No records

No readings yet

Scan your first strip to start your history.

Scan a strip

Loading

Loading your history…

Offline

No internet

You can keep scanning. Readings stay on this phone and send later.

No action needed

Saved locally

Saved on this phone

It will send by itself when you are online.

OK

Pending sync

Sending 2 readings

Keep the app open for a moment. You can still use it.

Updates on its own

Sync failed

Not sent

The connection dropped. Your reading is still on this phone.

Send again

Capture failed

Photo too blurry

Hold the phone still and keep the strip inside the frame.

Take it again

Uncertain reading

The phone isn't sure

Band 2 looks between two card colours. Please choose what you see.

Choose the band

Permission denied

Camera is off

MANTIKA CHECK needs the camera to photograph strips. Nothing else is recorded.

Open settings

Validation error

Oil temperature

1750

That looks too high. Frying oil is usually 150 to 200 °C. Or choose Don't know.

PJDSC 2026 · Team Innov

Research overview

Counts shown as [n] until data is collected

Two research tracks

1 · Public TPC dataset

Open →

Source	Li (2020), Mendeley Data
Temperatures	160 to 220 °C, 5 steps
Duration	0 to 30 h
Oil type	Not specified by the source

2 · Strip reader pilot

Open →

Consenting stalls	[n] · target 15 to 25
Repeat observation vendors	[n] · target 5 to 8
Usability sessions	[n] · min 5

Methodology progress

Step	Work	Status	Gate to pass
1 to 2	Download and audit Li (2020)	✓ Done	Units, rows, reuse terms checked
3	Forecasting experiment	--- In progress	MAE and RMSE on held out data
4	External validity write up	○ Not started	Domain shift documented
5 to 6	Field pilot and strip images	--- In progress	2 phones × 2 lighting settings
7	Reference TPC validation	■ Blocked	Needs a laboratory partner

Pilot area summary

Hidden until at least 5 stalls have consented, so no stall can be identified. Strip band counts, oil types and frying practices will appear here.

Manage participants

Limitations

- Published curves show a method, not vendor oil life
- Phone to human agreement is not TPC accuracy
- 25% TPC is a research benchmark, not Philippine law
- Pilot summaries hidden below 5 stalls

Deployment gates

1. Verify forecast errors on held out data
2. Confirm a lab method for calibration
3. No vendor countdown without local TPC data
4. Consent and pseudonymised records

Provenance

Li, Y. (2020). Frying oil data. doi:10.17632/4k7b395yxx.1

Yongyod and Kerdsin (2025), Foods 14(9), 1572

Track 1 · Methodology step 3

Machine learning laboratory

··· Experimental · never shown to vendors

Temperatures

☑ ○ 160 | ☑ □ 175 | ☑ △ 190 | ☑ ◇ 205 | ☑ × 220 held out

Model

Curve fit baseline | **Regularized regression**

Validation

Held out time | **Leave one temperature out**

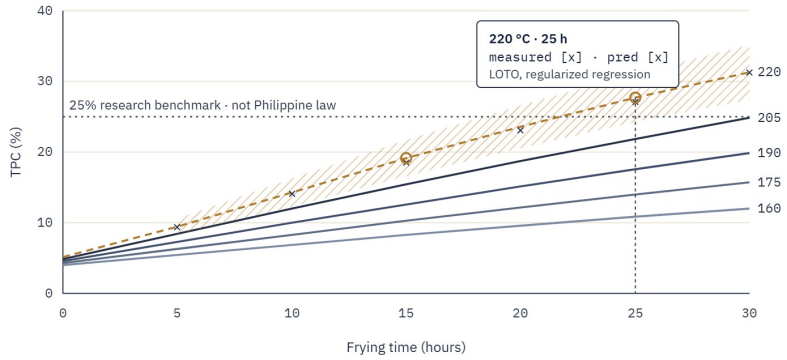
TPC over frying time

Total polar compounds, % of oil · illustrative curve shapes, not results

Show table

Measured, Li (2020) — Experimental prediction — Uncertainty range

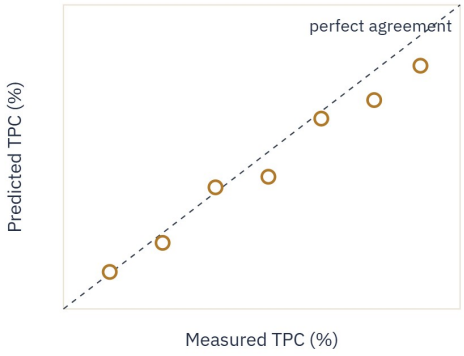
25% research benchmark



Source: Li (2020), Frying oil data, potato chip frying under controlled heat; oil type unspecified. Not Philippine vendor data.

Actual versus predicted

Held out 220 °C · illustrative



How to read it

Points on the dashed line are perfect predictions. Above the line, the model predicts more TPC than was measured.

Points shown

Only the held out series. Training points are never plotted here.

Error by model and validation

Model	Validation	MAE	RMSE
Curve fit	Held out time	not run	not run
Curve fit	LOTO	not run	not run
Regularized	Held out time	not run	not run
Regularized	LOTO	running	running

About this validation

Leave one temperature out trains on four series and predicts the fifth. It tests whether the model generalises across heat, not across oils or foods.

Time to 25% · research only

Temp	Crossing
160 °C	not yet determined
190 °C	not yet determined
220 °C	not yet determined

Reported only when observed or

Not a vendor forecast. Hours to 25% stay hidden from the mobile app until repeated local reference TPC measurements exist.

Track 1 · Methodology steps 1 and 2

Dataset explorer

Layout preview · example row states, values load from the CSV

total_polar_compoents.csv | acid_value.csv | carbonyl_value.csv

Temperature All five | **Hours** 0 to 30 | Only rows with issues | **Download filtered CSV**

Row	Temperature label	Time	TPC	Check
1	[label]	[h]	[%]	✓ OK
2	[label]	[h]	[%]	✓ OK
3	[label]	[h]	empty	▲ Missing value
4	[label]	[h]	[%]	◆ Unit to confirm
5	[label]	[h]	[%]	✓ OK

Rows 1 to 5 of [n] — Previous | Next

Columns

Column	Meaning	Type	Missing
[temperature]	Controlled frying temperature, one of five	category	[n]
[time]	Frying time; unit to be confirmed on import	number	[n]

Provenance

Author	Yumei Li (2020)
Host	Mendeley Data
DOI	10.17632/4k7b395yxx.1
Food	Potato chips
Oil	Not specified
Reuse terms	To verify before publication

Data quality checks

Missing values	not run
Duplicate rows	not run
Time units	not run
Temperature labels	not run
Measurement ranges	not run

Preprocessing rules

- TPC, acid and carbonyl values are never merged
- Rows from one temperature series are not treated as independent
- Every change is logged with the source row

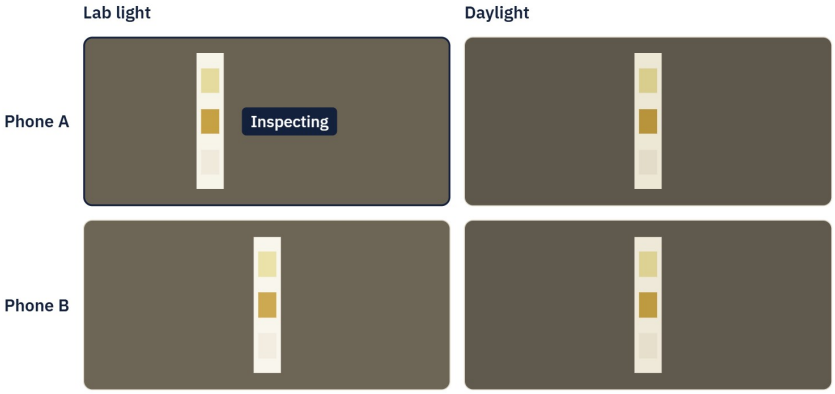
Track 2 · Methodology step 6

Strip laboratory S-014

← S-013 S-015 →

Same strip, four captures

All four stay in one data partition



Images are placeholders for real captures from the supervised campus laboratory.

Colour features · Phone A, lab light

Band	Sample	R G B	L*	a*	b*
1		[r g b]	[ ]	[ ]	[ ]
2		[r g b]	[ ]	[ ]	[ ]
3		[r g b]	[ ]	[ ]	[ ]

Person versus phone

Capture	Human reader	Phone, experimental	Agreement
Phone A · Lab	[band]	[band]	○ Pending
Phone A · Daylight	[band]	[band]	○ Pending
Phone B · Lab	[band]	[band]	○ Pending
Phone B · Daylight	[band]	[band]	○ Pending

Agreement, all strips

Across phones	not computed
Across lighting	not computed
Phone vs human	not computed

Two different questions

Camera to human agreement

Measured in this pilot. Does the phone read the band a person reads?

Strip to laboratory TPC

Not measured. Requires a qualified laboratory partner (validation gate 7).

Procedure

Strip method: Yongyod and Kerdsin (2025). Acidity strips are not substituted.

Track 2 · Methodology step 5

Vendor research management

Add participant

Search by ID, e.g. V-07

All | Consent given | Incomplete | Repeat observers

Participants · example records for layout

ID	Consent	Record	Strip readings	Repeat	
V-03	✓ Given	✓ Photo · ✓ Survey · ✓ Strip	[n]	Yes	Open
V-07	✓ Given	✓ Photo · ✕ Survey · ✓ Strip	[n]	No	Open
V-11	○ Pending	No data collected before consent	0	No	Record consent

Area summary is suppressed

Summaries appear when at least 5 stalls have consented. This prevents any single stall from being identified.

Privacy rules

- Only pseudonymous IDs appear here; names and stall locations are stored separately
- No reading is collected before consent
- Withdrawal removes the vendor's data from analysis
- Faces are never required in photos

Pilot scope

Consenting stalls	target 15 to 25
Repeat observers	target 5 to 8
Area	One, near campus

Outputs

Research reporting

Tables as CSV | Figures as SVG | Export report PDF

Include

- [x] Evaluation summary
- [x] Model comparison
- [x] Strip agreement
- [ ] Usability results (SUS)
- [x] Limitations · always included
- [x] Citations · always included

Reproducibility

Dataset	`Li 2020 v1`
Code	`[commit]`
Seed	`[seed]`
Generated	`on export`

MANTIKA CHECK

Preview · draft

Evaluated public data forecasting and a strip reader pilot

Team Innov · PJDSC 2026

1 · Evaluation summary

Question	Measure	Result
Forecast error, held out time	MAE, RMSE	pending
Forecast error, new temperature	MAE, RMSE	pending
Phone versus human reading	Agreement	pending

Results fill in only from completed runs. Nothing is estimated for the report.

2 · Model comparison

MAE and RMSE for both models, both validations, appear here after the runs finish.

Limitations

Published curves come from potato chip frying with an unspecified oil and demonstrate a method, not street vendor oil life. Phone strip readings are not direct TPC measurements; laboratory calibration remains future work. The 25% TPC line is a research benchmark, not Philippine law. The pilot does not detect pathogens or make regulatory rulings.

References

1. Li, Y. (2020). Frying oil data [Data set]. Mendeley Data. https://doi.org/10.17632/4k7b395yxx.1
2. Yongyod, R., & Kerdsin, A. (2025). Modified test kit for detecting polar compounds. Foods, 14(9), 1572.
3. Fan, Y., et al. (2021). Digital image colorimetry on smartphone. Measurement, 171, 108829.